

TIC 86956236

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 35.420027 d (ALIASED, see below)
-> corrected period: 70.840054 d
Epoch (t0) 1843.8893 BTJD
Duration 0.1200 d
Depth 4.974 % (49,742 ppm)
SNR 225.5

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

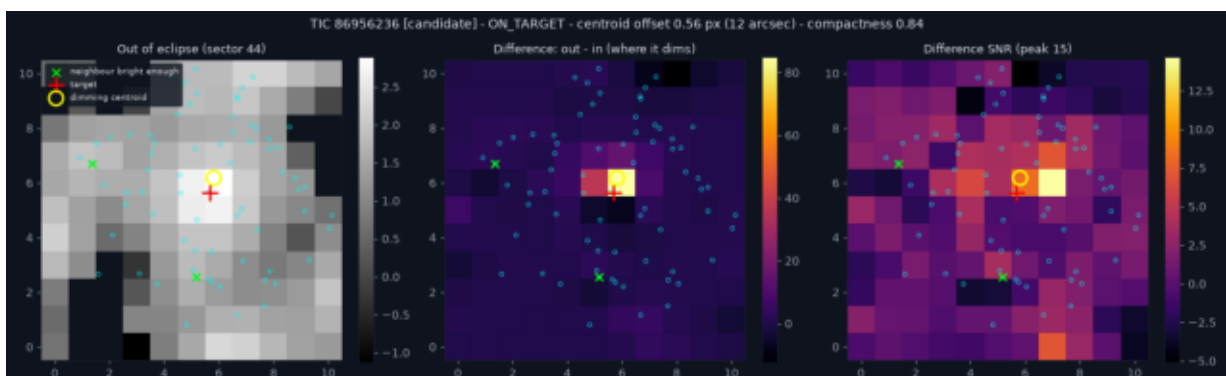
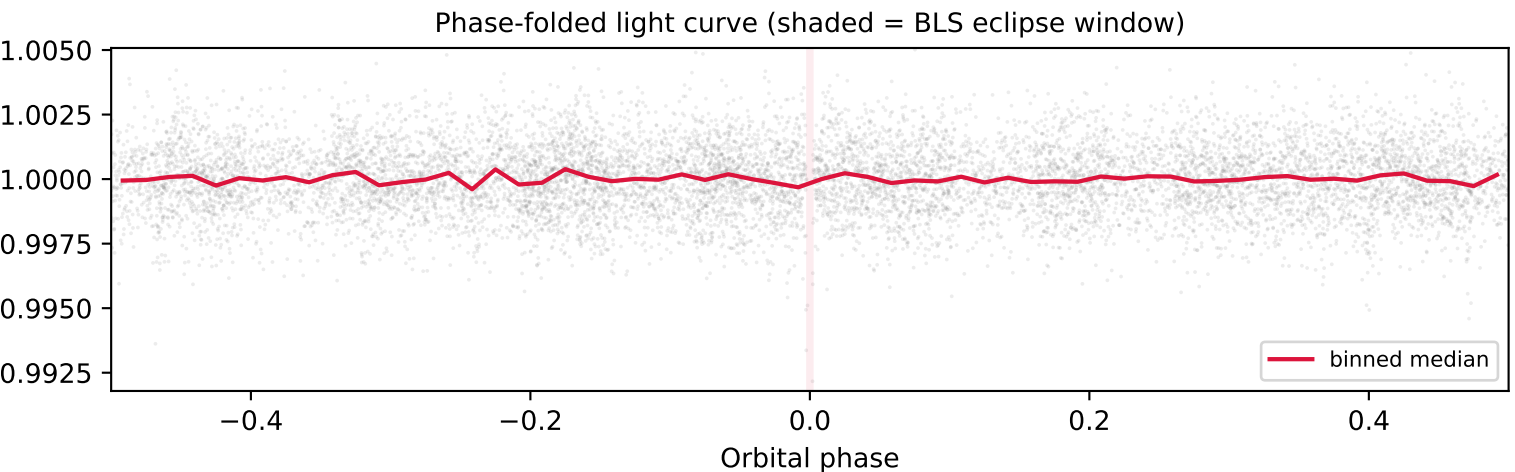
INDEPENDENT CHECKS (f3i gates)

Eclipses observed 3 (gate ≥ 3)
Robust depth ratio 1.07 (gate ≥ 0.50)
Robust depth sigma 200.3 (gate ≥ 7.0)
Odd/even z-score 6.26 (ALIAS FLAGGED)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 44 light-curve dip in this sector: 103.7 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 14.68 (gate ≥ 5.0) Compactness 0.84 (gate ≥ 0.50)
Dimming centroid offset 0.56 px (12 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 86956227
3.68 px away, dTmag +3.09 - farther from the dimming than the target



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 6.26 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 70.840054 d, in any write-up.